Harish Vijayakumar

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Education

Georgia Institute of Technology | Atlanta, GA *January 2026 – Present*
Bachelor of Science in Computer Engineering, GPA: 3.4/4.0
Relevant Coursework: Physical Foundations of Computer Engineering (Current), Programming for Hardware/Software Systems, Digital System Design, Circuit Analysis, Data Structures and Algorithms

University of Georgia (Transfer) | Athens, GA *August 2024 – December 2025*
Computer Systems Engineering, GPA: 3.75/4.0

Skills

Digital Design: RTL Design, Finite State Machines, Computer Architecture, FPGA Design
Languages: SystemVerilog, Verilog, VHDL, C/C++, Java, RISC-V Assembly, Python, MATLAB
Verification: SystemVerilog Testbenches, Coverage, Xcelium, SimVision, ModelSim, Waveform Debugging
Tools: Intel Quartus Prime, Cadence Innovus, Git, Linux
Hardware: Intel DE10-Lite FPGA, Arduino, Oscilloscopes, Logic Analyzer
Communication: Design proposals, technical reports, technical presentations, small-group instruction
Certifications: OSHA, CITI Human Subject Research
Organizations: SiliconJackets, GT Boxing Club (Secretary), Beta Chi Theta National Fraternity

Projects

8-Bit CPU | SystemVerilog, ModelSim, DE10-Lite (Personal) *June 2026 – July 2026*
Built a small CPU datapath to explore processor design and test it on an FPGA, with plans to expand the project.

- Created an 8-bit processor datapath implementing a program counter, instruction memory/decoder, register file, arithmetic logic unit (ALU), and FSM controller with 4 general-purpose registers meeting 50 MHz timing.
- Implemented five instructions (ADD, SUB, AND, OR, XOR) with register writeback and instruction execution.
- Verified RTL components using ModelSim and DE10-Lite hardware; utilized 70 registers and 150 logic elements, achieving 105 MHz Fmax, +10.5 ns setup slack, and 0 TNS.

FPGA UART Communication & DSP Platform | SystemVerilog, DE10-Lite (Personal) *May 2026 – June 2026*
Developed a communication and DSP platform to deepen my understanding of Verilog through hands-on design.

- Designed and implemented a modular SystemVerilog UART communication platform featuring UART TX/RX, FIFO buffering, command parsing, and FSM controller, supporting 115200 baud and meeting 50 MHz timing.
- Developed a DSP subsystem including a moving average filter, modifiable clock divider, and 7-segment display driver; utilized 503 logic elements and 282 registers, achieving 125 MHz Fmax and +12.0 ns setup slack.
- Verified RTL in ModelSim, then tested on hardware over a CP2102 USB-to-UART link with the Arduino IDE.

Experience

RevSync - Startup | Lead Hardware Engineer *January 2026 – Present*
Vehicle intelligence startup combining vehicle telemetry, lap timing, and AI-powered coaching in a modular platform for drivers and motorsports teams. *Atlanta, GA*

- Contributed to initial design and hardware requirements for an OBD plug-in module that turns vehicle data into AI driving feedback, choosing the vehicle data the module will capture.
- Co-presented RevSync at the GT Klaus Startup Challenge, communicating technical direction and roadmap, achieving acceptance to CREATE-X LAUNCH.

SiliconJackets | ASIC Calculator Chip (Individual - onboarding) | Cadence Tools *January 2026 – March 2026*
Georgia Tech student organization focused on chip design and student-led tapeouts, with hands-on work in RTL design, verification, and physical design. *Atlanta, GA*

- Implemented a SystemVerilog 64-bit arithmetic calculator with SRAM-based operand storage and FSM, combining upper and lower 32-bit operations with carry propagation.
- Verified the design in Xcelium and SimVision with a driver, monitor, sequencer, scoreboard, 3 assertions, and 400 constrained-random transactions, achieving 99.95% coverage and 100% assertion coverage.

- Resolved physical design rule check (DRC) violations in Cadence Innovus by reordering metal routing layers, achieving a DRC-clean final layout.

University of Georgia | Peer Learning Assistant *August 2025 – December 2025*

Intro to Electrical & Computer Systems Engineering — Introduces core ECE concepts and builds technical skills through lectures, design projects, and labs. *Athens, GA*

- Supported 50+ students in mastering core engineering concepts through collaborative learning and guided problem-solving.
- Led small-group discussions and provided individualized guidance to build students' confidence in applying engineering principles to real-world problems.

University of Georgia - Cultivate Lab | Athens, GA *January 2025 – December 2025*

Undergraduate Researcher

Interdisciplinary research lab focused on engineering education, mental health, and student retention.

- Screened 5,000+ peer-reviewed articles using Rayyan to extract and synthesize data from 35 qualifying studies, supporting a scoping review on STEM graduate mental health.
- Authored and presented original research findings at Frontiers in Education 2025 and The American Society for Engineering Education 2026 conferences.

PC Build and Repair Service | Independent Venture *January 2014 – Present*

Technician

Founded and operated a self-directed computer hardware and electronics service supporting individual and small business clients.

- Assembled 40+ custom PCs and repaired 100+ devices including desktops, laptops, and peripherals.
- Diagnosed hardware and software failures across Intel, AMD, and NVIDIA platforms.

Publications

[1] J. Czworka, H. Vijayakumar, S. T. Ige, T. R. Feyijimi, and S. J. Bork, "WIP: A scoping literature review on survey instruments used to examine STEM graduate students' mental health experiences," in *Proc. IEEE Frontiers Educ. Conf. (FIE)*, 2025, doi: 10.1109/FIE63693.2025.11328435.

[2] N. Cortez, H. Vijayakumar, E. B. Strehl, S. J. Bork, and A. W. Johnson, "WIP: Using photoelicitation to explore coping strategies of first-generation and low-SES engineering students," in *Proc. ASEE Annu. Conf. & Expo.*, Charlotte, NC, USA, 2026, doi: 10.18260/1-2–60816.